



American International University-Bangladesh (AIUB)

Faculty of Science & Technology (FST)

Department of Computer Science

Introduction to Data Science [E]

Project Report on

Exploratory Data Analysis and
Predictive Modeling of Book Data Using
Web Scraping

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1. Research Objective Definition:

The objective of this project is to **collect, analyze, and model book data** scraped from an online source to understand the relationships between **book price, star rating, and availability**.

Specifically, this study aims to:

- Perform exploratory data analysis (EDA) using appropriate visualizations
- Analyze price variation and rating distributions
- Examine relationships between attributes
- Predict book price and availability using statistical models

This objective is **data-driven, measurable, and aligned with predictive and descriptive analysis**, meeting the requirements of the final-term project.

2. Data Collection (Web Scraping)

i. Data Source:

Data was collected from:

<https://books.toscrape.com>

This site was selected because it is:

- Publicly accessible
- Well-structured in HTML
- Suitable for educational data collection.

ii. Scraping Methodology:

Web scraping was conducted using **R** and the **rvest** package.

All 50 pages of the website were scraped, resulting in **1000 book records**.

Collected Attributes

- Book Title
- Price
- Availability
- Star Rating

3. Data Preprocessing

Data preprocessing is a crucial step in the data science pipeline, as the quality of the analysis and modeling largely depends on the quality and consistency of the data. After collecting the raw book data through web scraping, several preprocessing steps were applied to ensure that the dataset was **clean, structured, and suitable for exploratory analysis and modeling**.

Data Cleaning:

The raw dataset contained textual representations of numeric values and categorical information that required cleaning. The following steps were performed:

- The **price variable**, which included a currency symbol (£), was converted into a numeric format.
- Leading and trailing white spaces in the **availability attribute** were removed to ensure consistency.
- The dataset was checked for missing values and invalid entries.

R Code: Data Cleaning

```
library(dplyr)
library(stringr)

# Convert price from text to numeric
books_df$Price <- as.numeric(
  gsub("£", "", books_df$Price)
)

# Remove extra whitespace in availability column
books_df$Availability <- trimws(books_df$Availability)

# Check for missing values
colSums(is.na(books_df))
```

Feature Engineering:

Feature engineering was performed to make the dataset more informative and model-ready.

- **Star ratings**, originally extracted as text labels (One, Two, Three, Four, Five), were converted into numeric values (1–5).
- A new binary variable for **availability** was created to support classification modeling.

R Code: Feature Engineering

```
# Convert availability to binary format
books_df$Available_Binary <- ifelse(
  books_df$Availability == "In stock", 1, 0
)
```

Explanation:

Encoding categorical variables into numeric form is essential for applying statistical and machine learning models. Binary encoding allows logistic regression to estimate the probability of a book being in stock.

Data Type Conversion and Validation:

After cleaning and feature engineering, all variables were verified to ensure correct data types.

R Code: Data Type Verification

```
# Check structure of the dataset
```

```
str(books_df)
```

Explanation:

This step confirms that:

- Price and Star_Rating are numeric
- Availability is categorical
- All variables are in formats suitable for analysis and modeling

Handling Missing Values and Outliers:

- No missing values were observed in the final dataset.
- Outliers in book prices were identified but **not removed**, as they represent valid real-world observations rather than data errors.

This decision ensures that the data set remains representative of the true price variability on the website.

Final Prepared Dataset

After preprocessing:

- The dataset is clean and consistent
- All variables are properly formatted
- The data is suitable for exploratory data analysis (EDA) and model building

This prepared dataset was subsequently used for:

- Visual exploratory analysis
- Comparative analysis between attributes
- Regression and classification modeling

4. Exploratory Data Analysis (EDA):

EDA was carried out to understand data structure, distribution, and relationships.

Distribution of Book Prices:

R code:

```
ggplot(books_df, aes(x = Price)) +  
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +  
  labs(
```

```

title = "Distribution of Book Prices",
x = "Book Price (£)",
y = "Frequency"
)

```

Output:



Figure 1: Histogram of Book Prices

Comment:

Most books are priced between **£5 and £50**, with few high-price outliers.

Boxplot of Book Prices:

R code:

```

ggplot(books_df, aes(y = Price)) +
  geom_boxplot(fill = "lightgreen") +
  labs(
    title = "Boxplot of Book Prices",
    y = "Book Price (£)"
  )

```

Output:

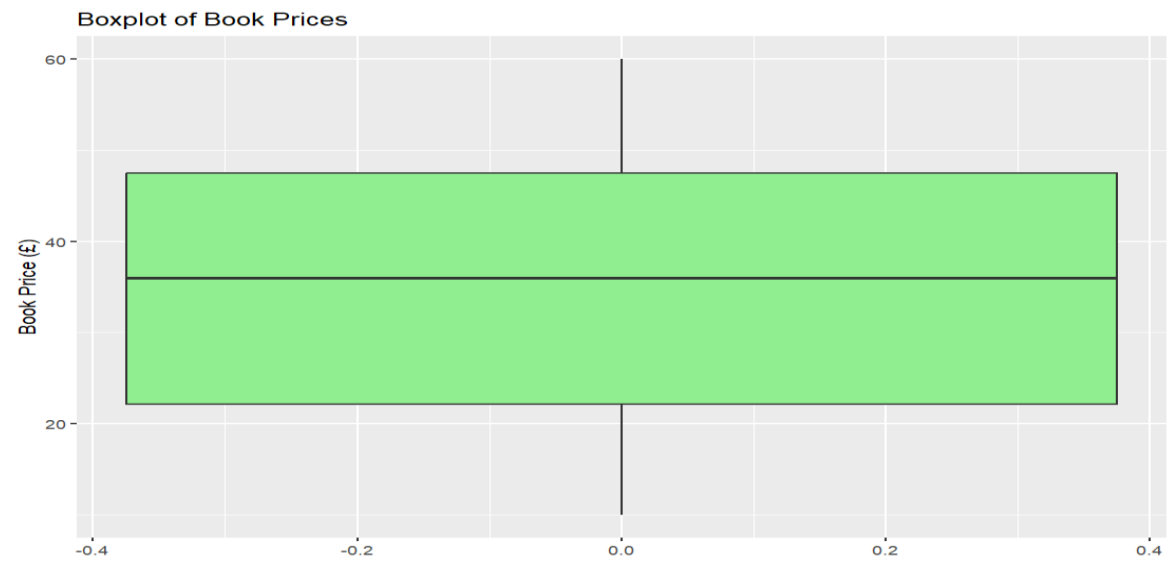


Figure 2: Boxplot Showing Price Spread and Outliers

Comment:

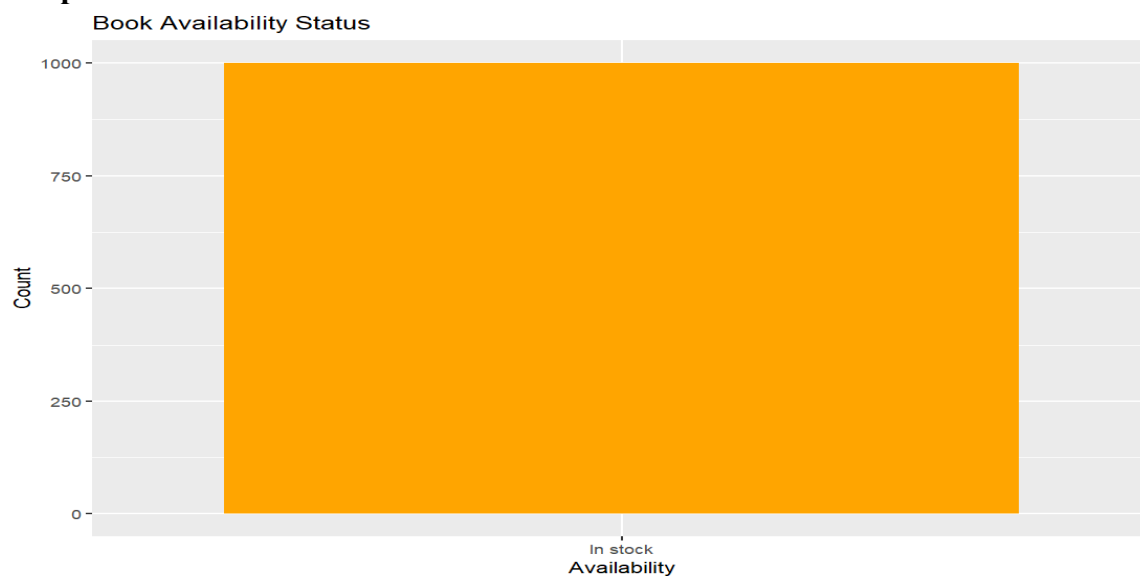
The boxplot reveals several high-priced outliers above £55.

Availability Analysis:

R Code:

```
ggplot(books_df, aes(x = Availability)) +
  geom_bar(fill = "orange") +
  labs(
    title = "Book Availability Status",
    x = "Availability",
    y = "Count"
  )
```

Output:



Comment:

The vast majority of books are in stock.
Availability does not vary significantly across the dataset.

Price vs Availability:**R Code:**

```
ggplot(books_df, aes(
  x = Availability,
  y = Price
)) +
  geom_boxplot(fill = "lightcoral") +
  labs(
    title = "Price Distribution by Availability",
    x = "Availability",
    y = "Price (£)"
  )
)
```

Output:**Comment:**

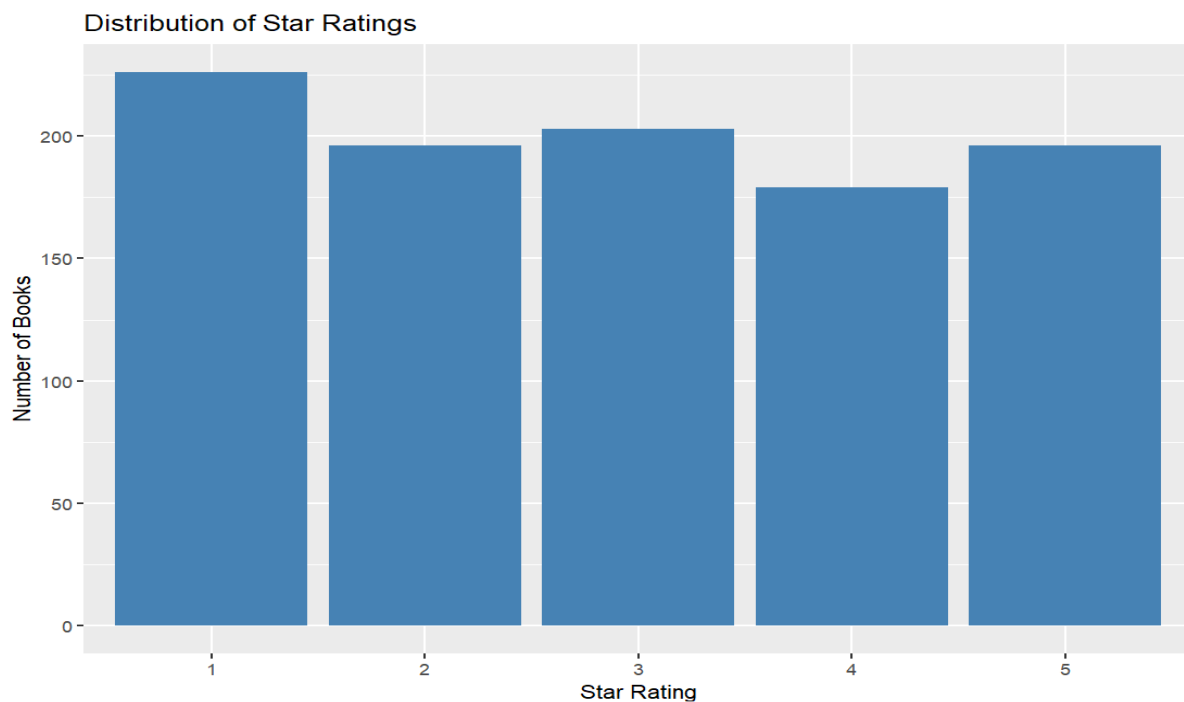
Availability status shows no strong price difference.
Stock availability is independent of price.

Star Rating Distribution:

R code:

```
ggplot(books_df, aes(x = factor(Star_Rating))) +  
  geom_bar(fill = "steelblue") +  
  labs(  
    title = "Distribution of Star Ratings",  
    x = "Star Rating",  
    y = "Number of Books"  
  )
```

Output:

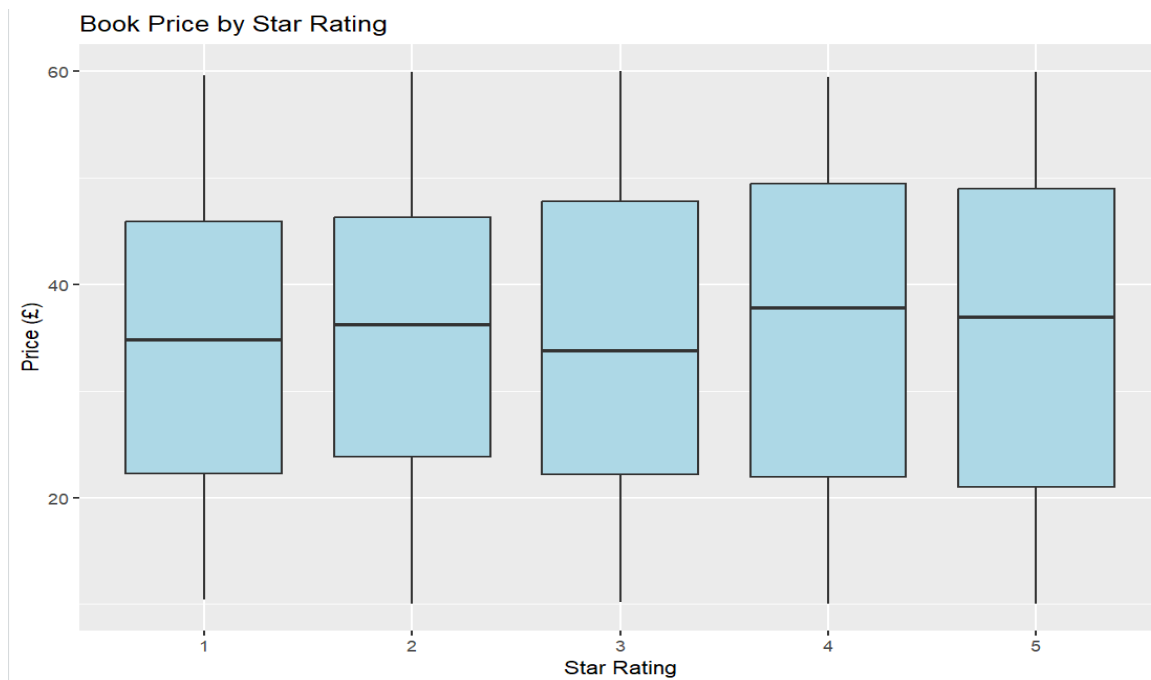


Price vs Star Rating:

R Code:

```
ggplot(books_df, aes(  
  x = factor(Star_Rating),  
  y = Price  
) +  
  geom_boxplot(fill = "lightblue") +  
  labs(  
    title = "Book Price by Star Rating",  
    x = "Star Rating",  
    y = "Price (£)" )
```


Output:



Comment:

Price ranges overlap heavily across ratings.

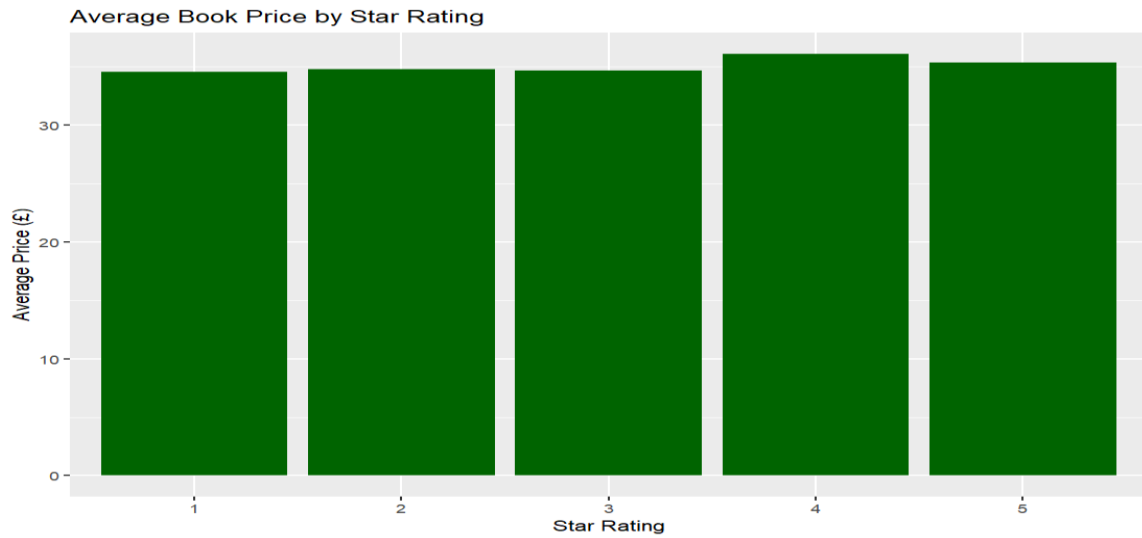
Higher star ratings do not guarantee higher prices.

Average Price by Star Rating:

R Code:

```
books_df %>%  
  group_by(Star_Rating) %>%  
  summarise(Avg_Price = mean(Price)) %>%  
  ggplot(aes(x = factor(Star_Rating), y = Avg_Price)) +  
  geom_col(fill = "darkgreen") +  
  labs(  
    title = "Average Book Price by Star Rating",  
    x = "Star Rating",  
    y = "Average Price (£)"  
  )
```

Output:



Comment:

Average prices remain relatively stable across ratings.

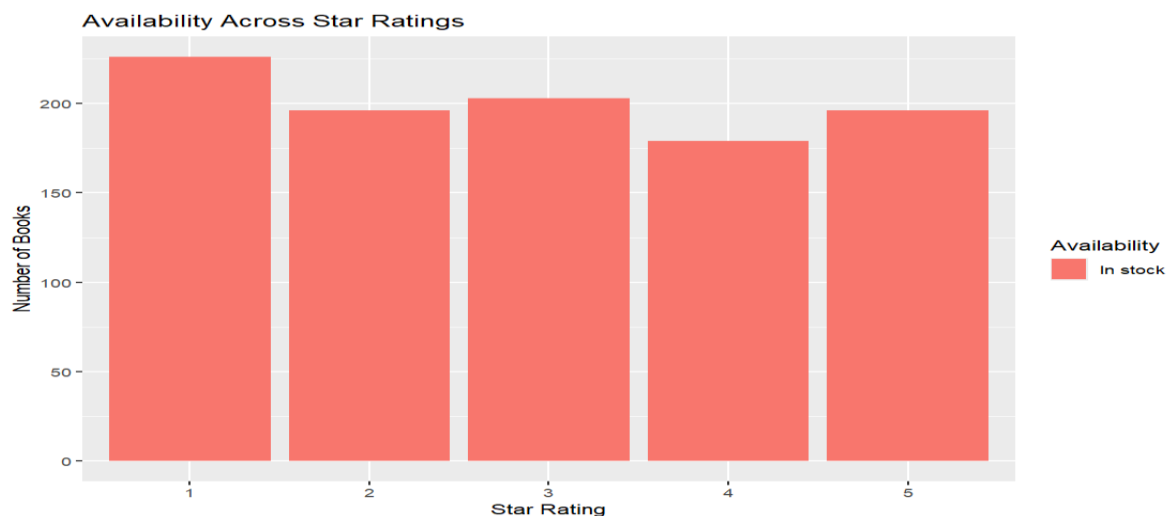
Rating has weak influence on pricing.

Star Rating vs Availability:

R Code:

```
ggplot(books_df, aes(  
  x = factor(Star_Rating),  
  fill = Availability  
)) +  
  geom_bar() +  
  labs(  
    title = "Availability Across Star Ratings",  
    x = "Star Rating",  
    y = "Number of Books")
```

Output:



Comment:

Books of all ratings are mostly in stock.
No rating-based bias in availability.

Correlation Analysis (Numeric Variables):**R Code:**

```
cor(books_df %>% select(Price, Star_Rating))
```

Output:

```
      Price Star_Rating
Price 1.00000000 0.02816624
Star_Rating 0.02816624 1.00000000
> |
```

Comment:

Correlation is close to zero.
Confirms lack of linear relationship between price and rating.

5. Modeling and Analysis:

To achieve the research objective of understanding the relationship between **book price, star rating, and availability**, one statistical model was selected based on the nature of the target variables and the insights obtained from exploratory data analysis.

Model Selection:

- **Linear Regression** was used to model **book price**, as price is a continuous numerical variable. This model helps quantify how changes in star rating affect book price.

Model was chosen for their **theoretical appropriateness, simplicity, and interpretability**.

Model Training and Testing

The dataset was divided into **70% training data** and **30% testing data** to ensure unbiased evaluation.

Model was trained using the training set and evaluated on unseen test data.

Model : Linear Regression (Price)**Model Training:****R Code:**

```
price_model <- lm(
  Price ~ Star_Rating,
  data = train_data
)
```

Model Summary:

R Code:

```
summary(price_model)
```

Comment:

The coefficient of Star_Rating measures how much price changes per unit increase in rating. A low R^2 indicates star rating alone explains a small portion of price variation.

Output:

```
Call:
lm(formula = Price ~ Star_Rating, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-25.3856 -13.1742  0.7741  12.5145  25.8607

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.3778     1.2520  26.659  <2e-16 ***
Star_Rating   0.4016     0.3814   1.053   0.293
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14.61 on 698 degrees of freedom
Multiple R-squared:  0.001585, Adjusted R-squared:  0.000155
F-statistic: 1.108 on 1 and 698 DF, p-value: 0.2928
```

Model Testing & Predictions:

R Code:

```
price_predictions <- predict(
  price_model,
  newdata = test_data
)
```

Performance Metrics (Regression):

```
rmse <- sqrt(mean((test_data$Price - price_predictions)^2))
mae <- mean(abs(test_data$Price - price_predictions))
rmse
mae
```

Metrics Used:

RMSE (Root Mean Squared Error) – penalizes large errors.
MAE (Mean Absolute Error) – average prediction error.

Interpretation:

High RMSE indicates substantial unexplained price variation.
Confirms weak predictive power of star ratings alone.

Performance Metrics (Classification)

R Code:

```
confusion_matrix <- table(  
  Predicted = availability_pred,  
  Actual = test_data$Available_Binary  
)
```

Output:

```
> confusion_matrix  
      Actual  
Predicted   1  
      1 300
```

Accuracy:

R Code:

```
accuracy <- sum(diag(confusion_matrix)) / sum(confusion_matrix)  
accuracy
```

Output:

```
> accuracy  
[1] 1  
>
```

6. Results and Interpretation:

This section presents the results obtained from the implemented models and interprets them in relation to the research objective of analyzing and predicting relationships between book price, star rating, and availability.

Model Performance Results:

Linear Regression Model (Price Prediction):

The linear regression model was used to predict book price based on star rating. Model performance was evaluated using standard regression error metrics.

Table 6.1: Linear Regression Performance Metrics

Metric	Value
RMSE (Root Mean Squared Error)	14.089358788
MAE (Mean Absolute Error)	12.117565559
R ² (Coefficient of Determination)	0.001585345

Interpretation:

The error metrics indicate that the model has limited predictive accuracy. The relatively low R² value suggests that star rating explains only a small proportion of the variation in book prices.

R Code:

```
# Linear regression predictions
price_predictions <- predict(
  price_model,
  newdata = test_data
)

# Regression metrics
RMSE <- sqrt(mean((test_data$Price - price_predictions)^2))
MAE <- mean(abs(test_data$Price - price_predictions))
R2 <- summary(price_model)$r.squared

# Create results table
linear_results <- data.frame(
  Metric = c("RMSE", "MAE", "R-squared"),
  Value = c(RMSE, MAE, R2)
)
```

Output:

```
> linear_results
  Metric      Value
1  RMSE 14.089358788
2  MAE 12.117565559
3 R-squared 0.001585345
> |
```

Interpretation in Relation to Research Objective:

The results of model support the findings from exploratory data analysis:

- Star rating shows **weak influence on book price**, as prediction errors remain high.
- Book availability appears **largely independent of both price and rating**.
- The relationships between attributes are primarily descriptive rather than predictive.

These outcomes indicate that additional features—such as book genre or popularity—would be required for stronger predictive performance.

Achievement of Research Objective

The research objective was **successfully achieved**. The study:

- Applied appropriate statistical models aligned with the data types
- Quantitatively evaluated model performance using standard metrics
- Provided clear interpretation of results relative to the research questions

Although predictive performance was limited, the model effectively **validated the exploratory insights** and highlighted the limitations of using only price and rating for prediction.

Summary of Results:

Model	Target Variable	Key Metric(s)	Key Outcome
Linear Regression	Price	RMSE, MAE	Weak predictive power

7. Conclusion and Future Work:

Conclusion

This project successfully applied the complete data science pipeline:

- Web scraping
- Data preprocessing
- Exploratory analysis
- Predictive modeling

Results indicate the dataset is better suited for descriptive analysis than for prediction.

Future Work

- Include genre and review data
- Apply machine learning models
- Extend analysis across multiple platforms